



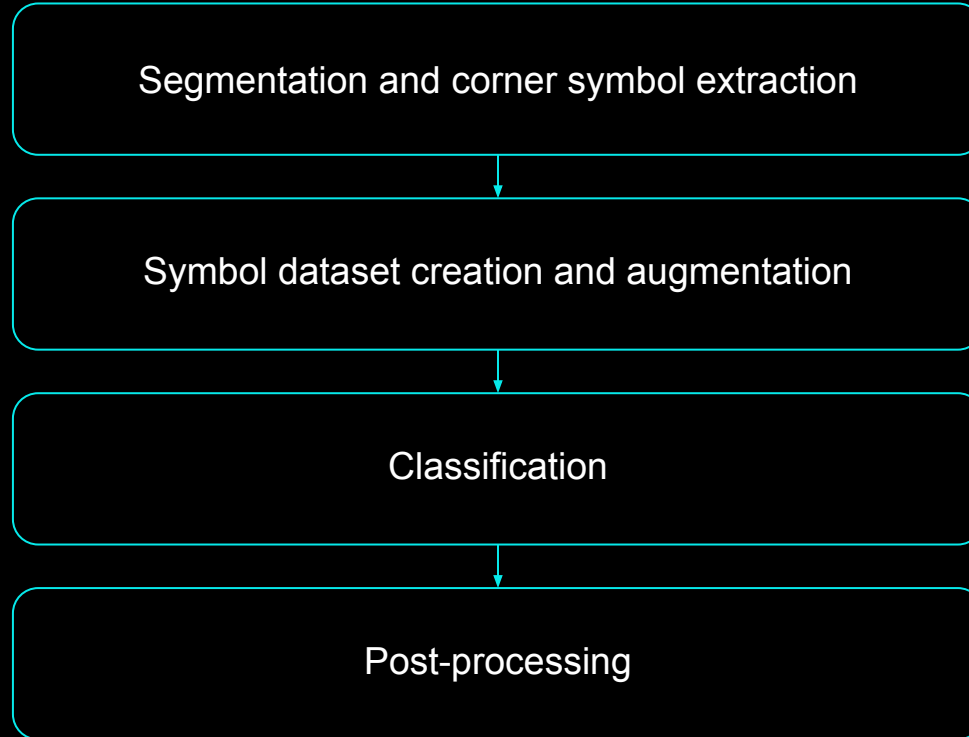
UNO Vision Challenge

Image Analysis and Pattern Recognition



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The Pipeline



Segmentation: the Input

L1000995.jpg



L1000896.jpg



Segmentation: Token Detection

L1000995.jpg: yellow token



L1000896.jpg: grey token



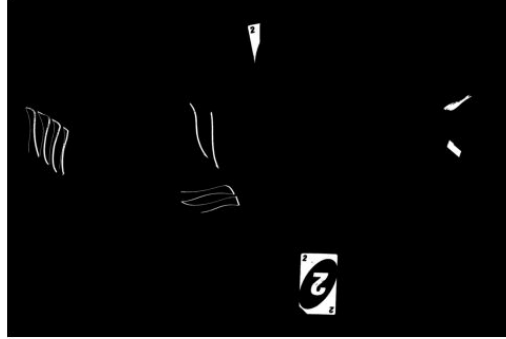
Segmentation: Color Thresholding

Raw color masks after area filtering

Blue



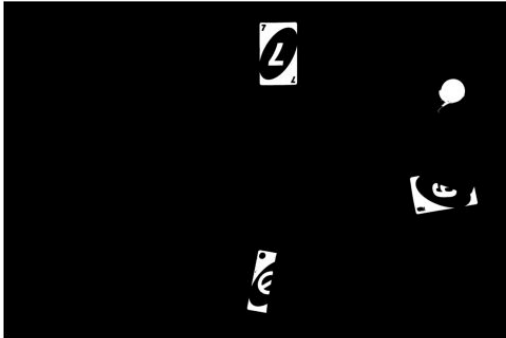
Green



Red



Yellow



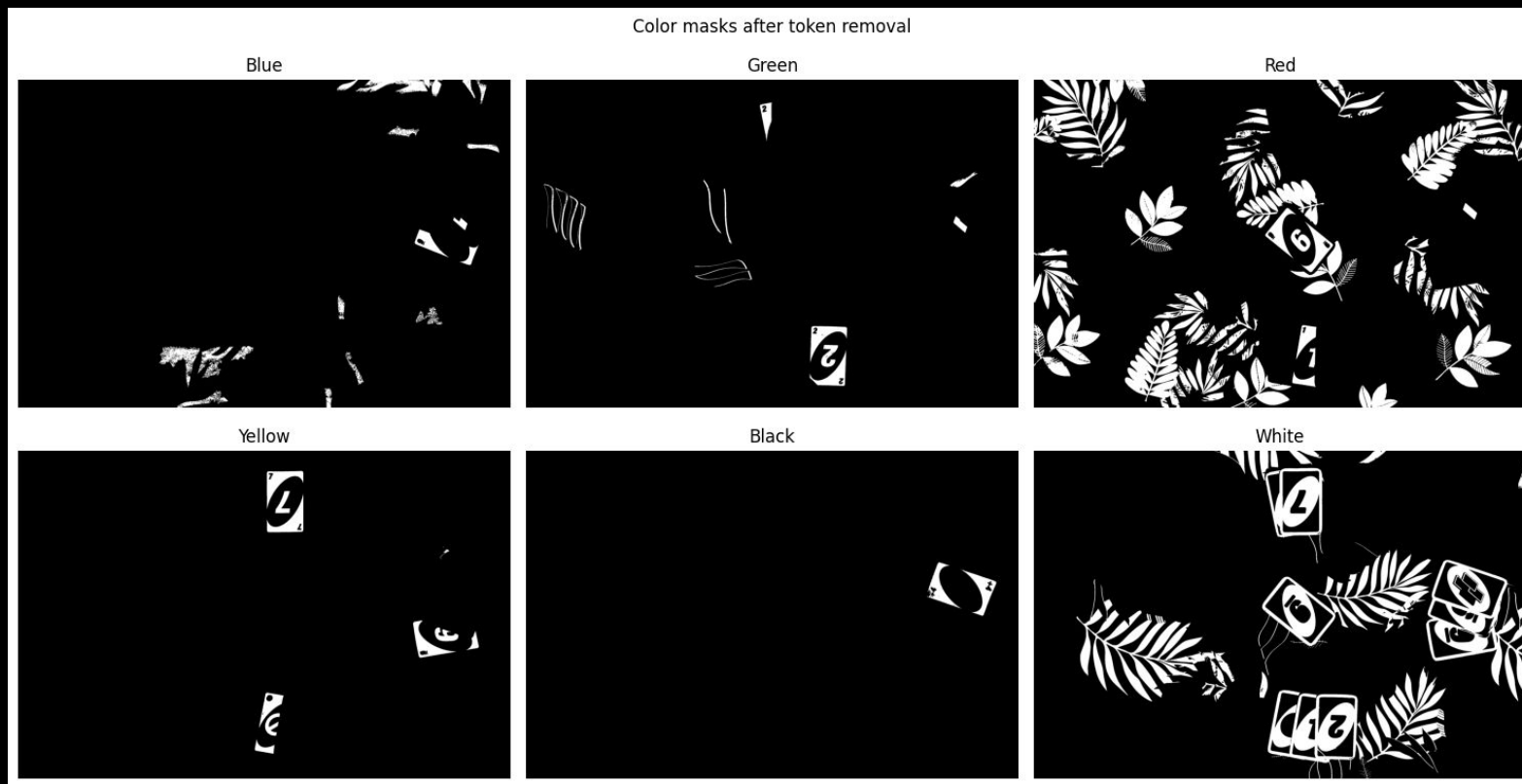
Black



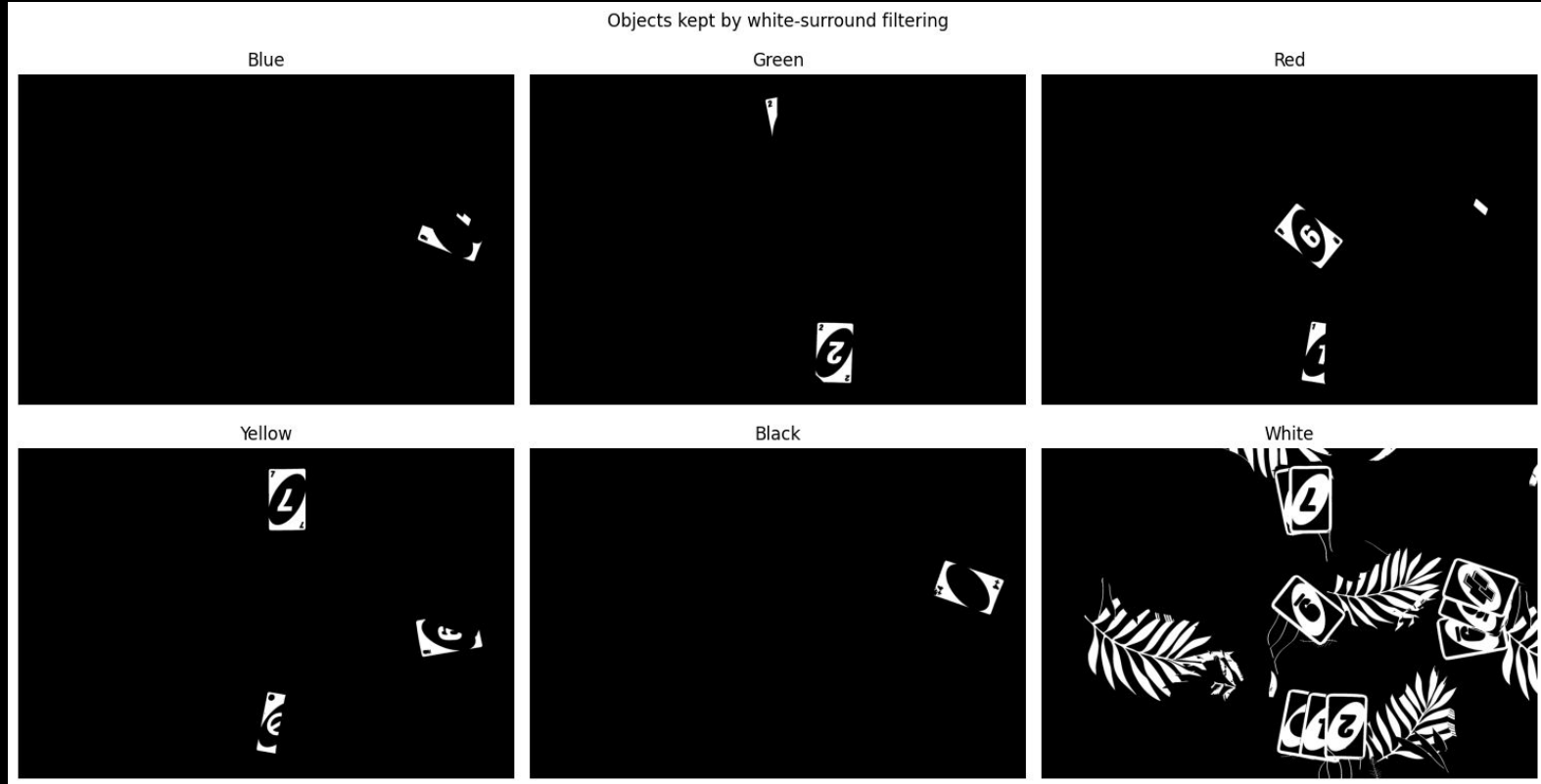
White



Segmentation: Token Removal



Segmentation: Filtering out non-card objects

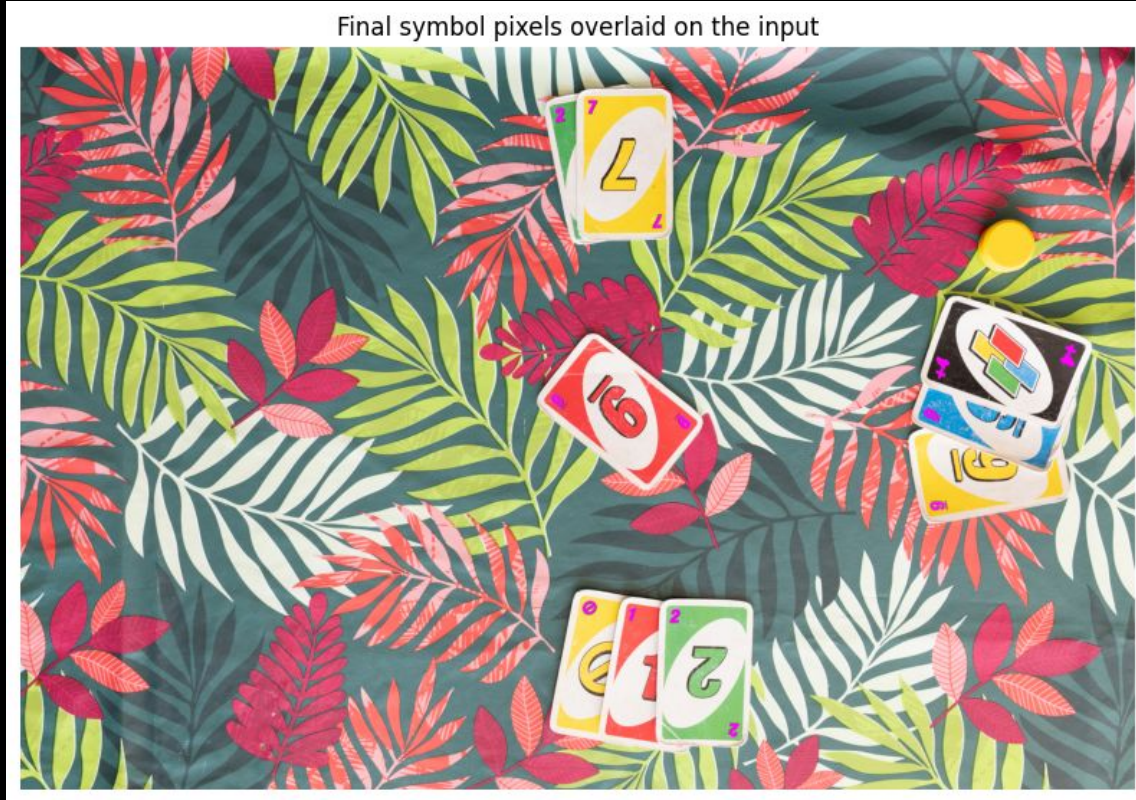


Segmentation: Finding symbols

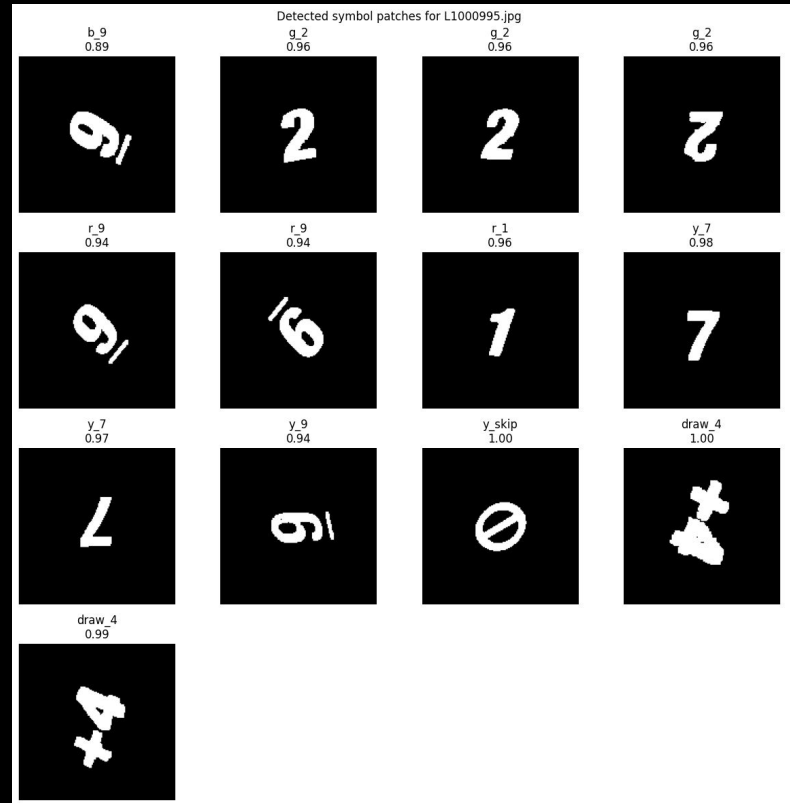
All holes inside colored card regions



Segmentation: Filtering out non-symbol objects



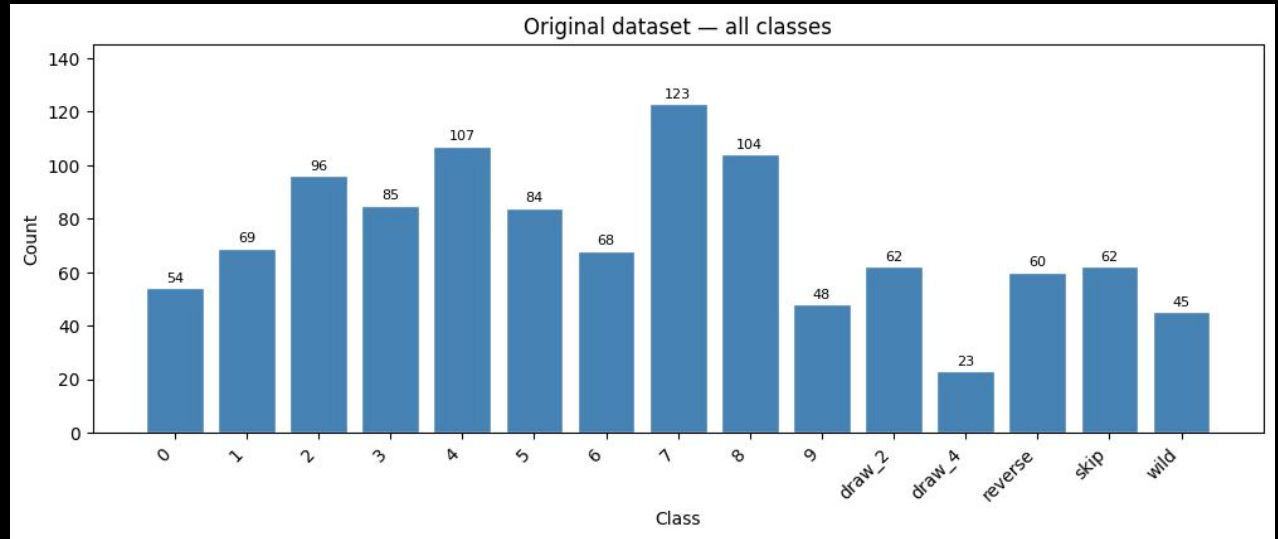
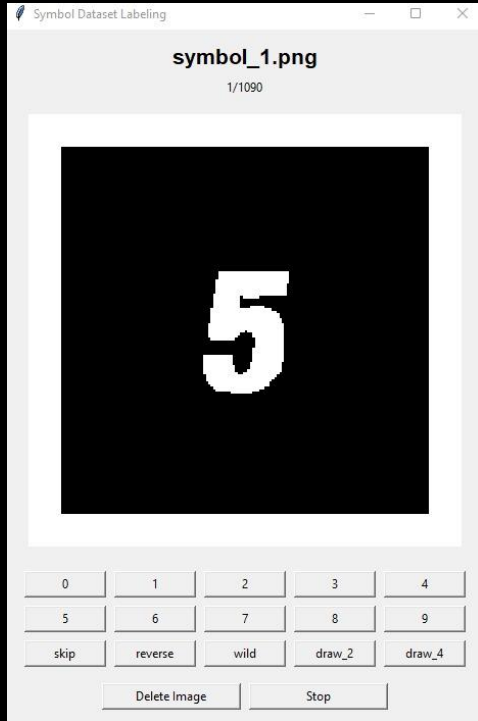
Segmentation: Symbol extraction



Segmentation: Limitations

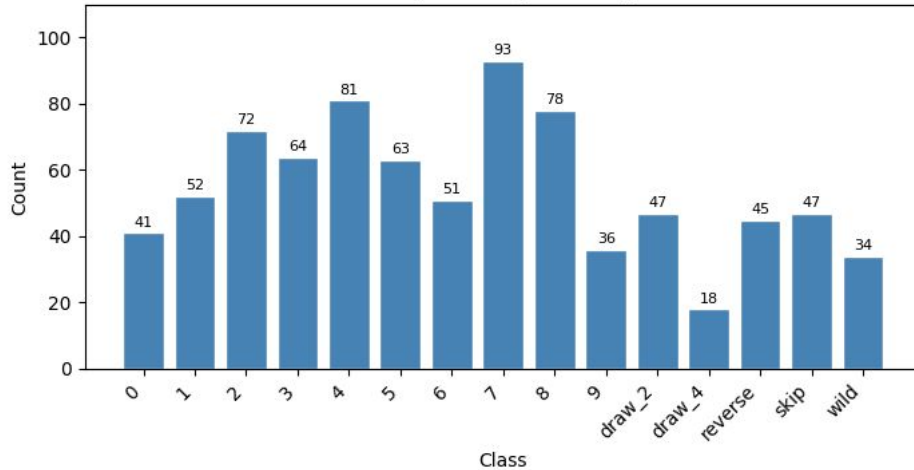


Dataset Preparation for Classification: Building and Labeling

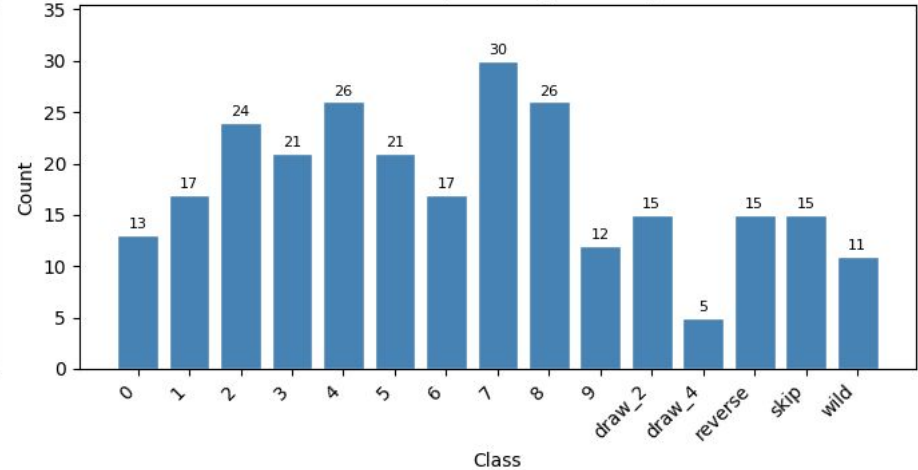


Dataset Preparation for Classification: Train & Validation Split

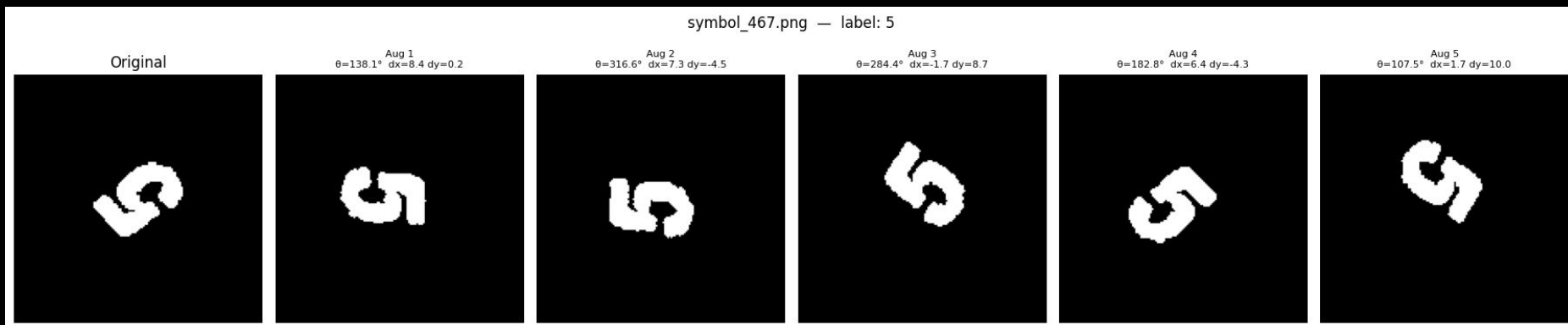
Train split — original images



Validation split — original images

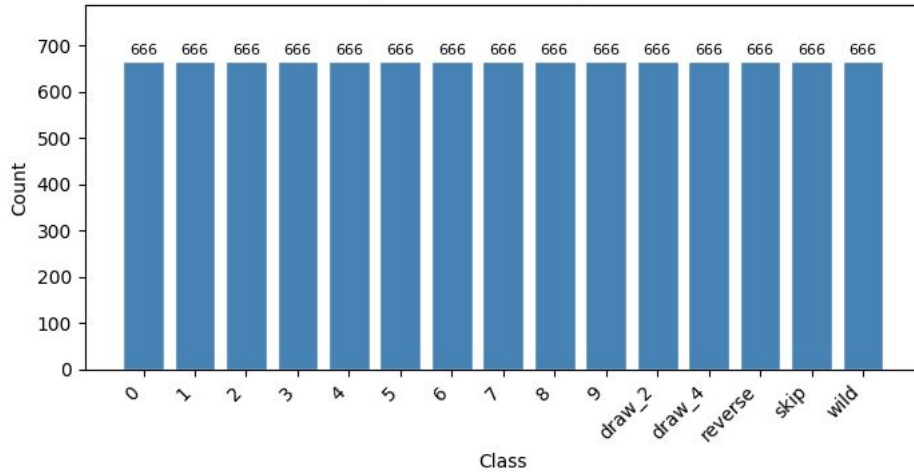


Dataset Preparation for Classification: Augmentation

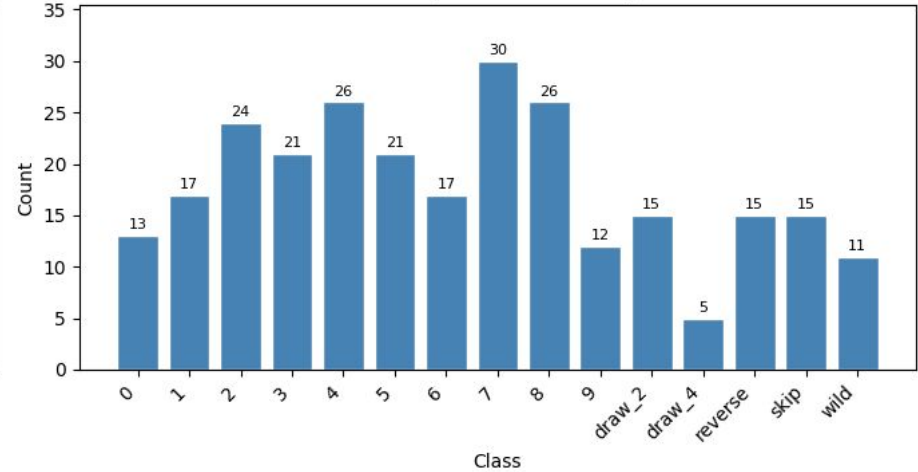


Dataset Preparation for Classification: Final Train & Validation Split

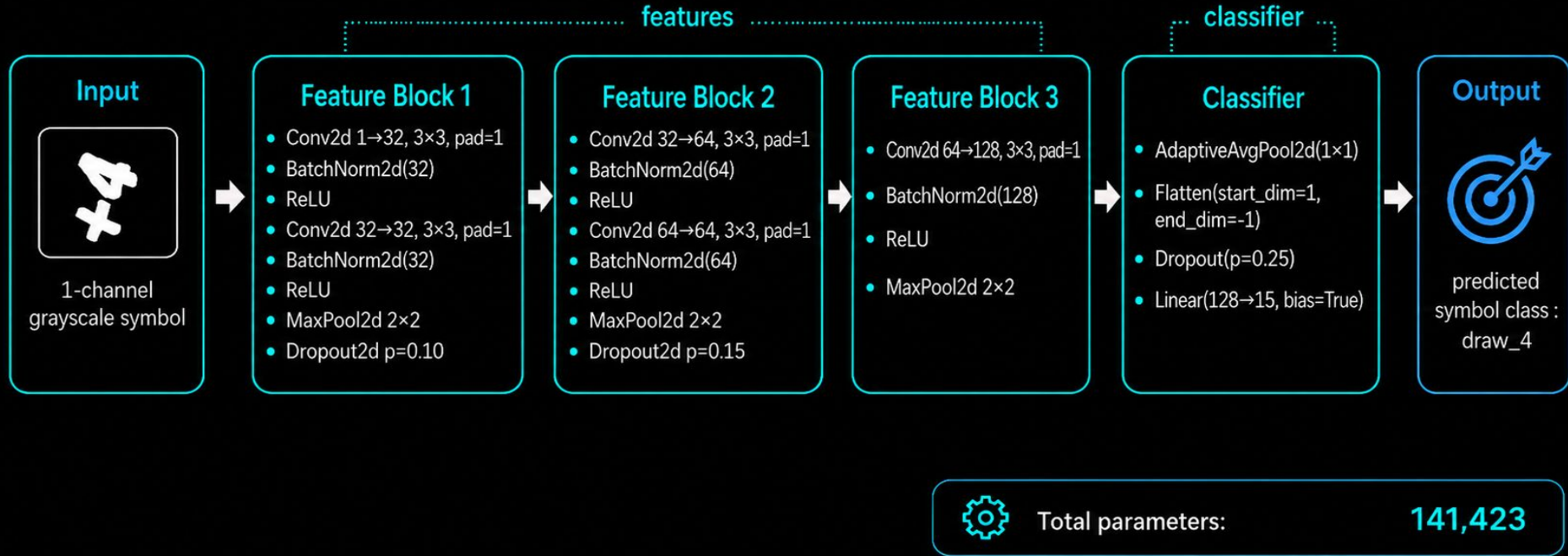
Train split - final after augmentation



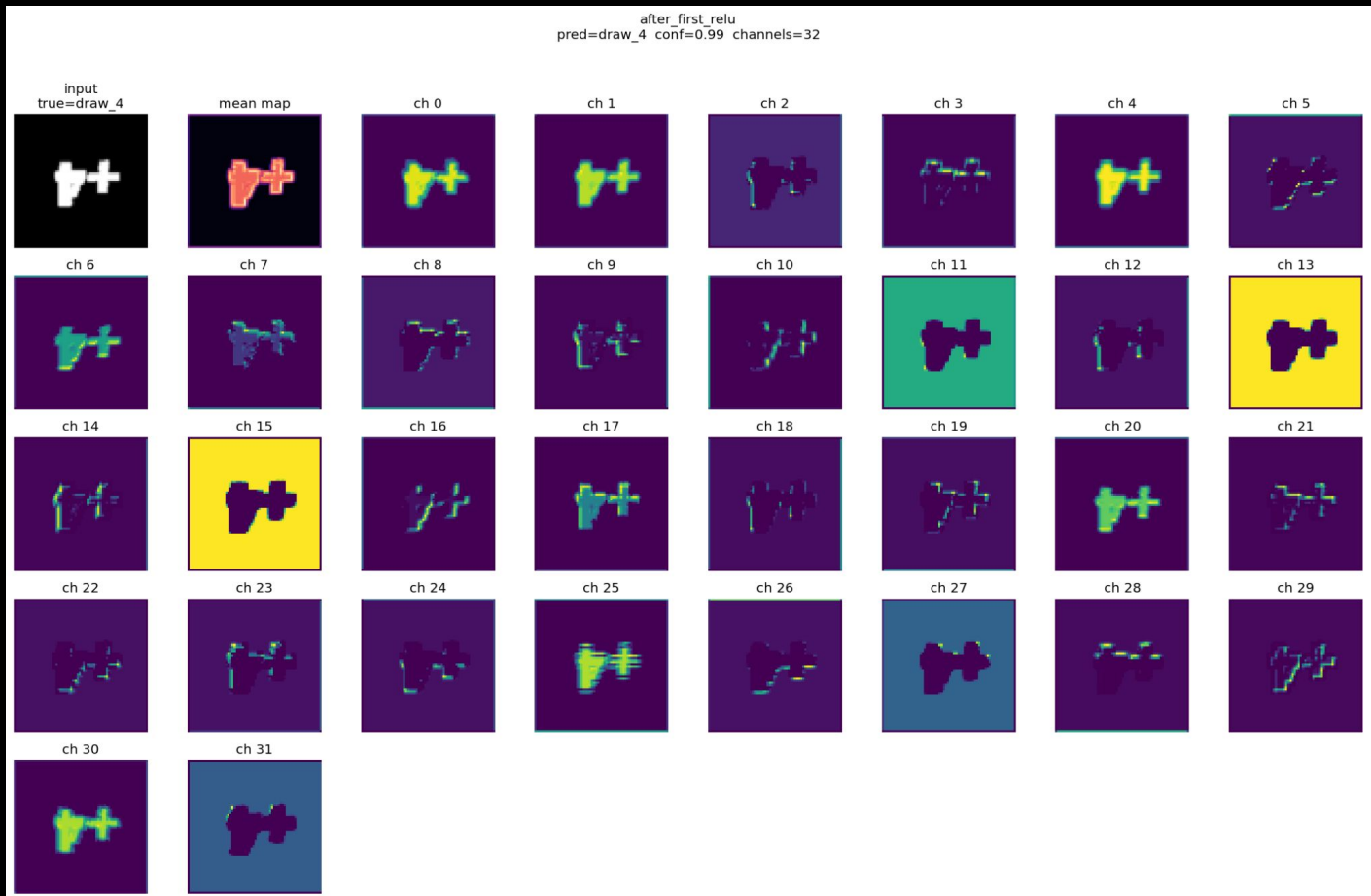
Validation split - held-out originals



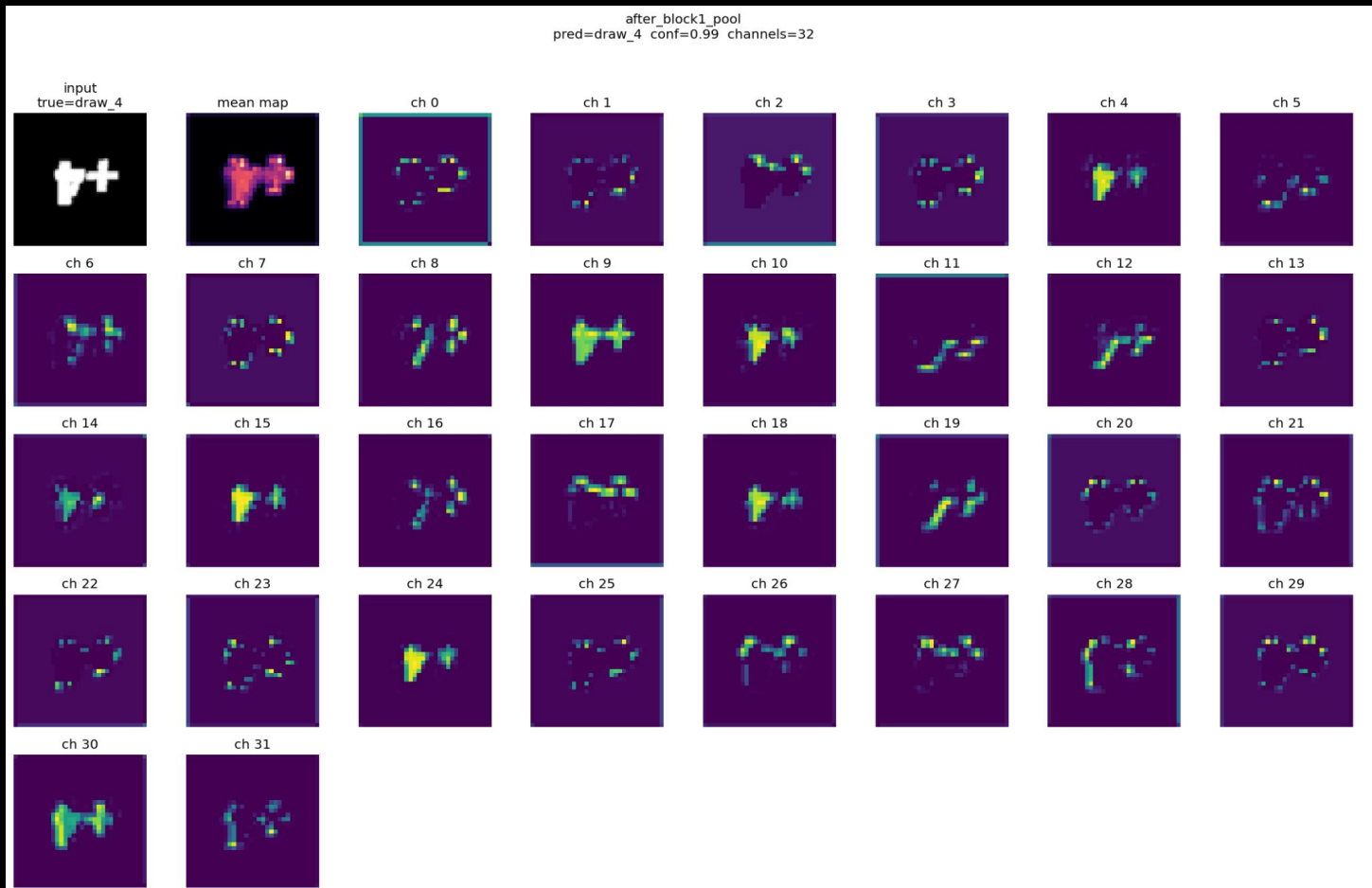
Classification: CNN Architecture



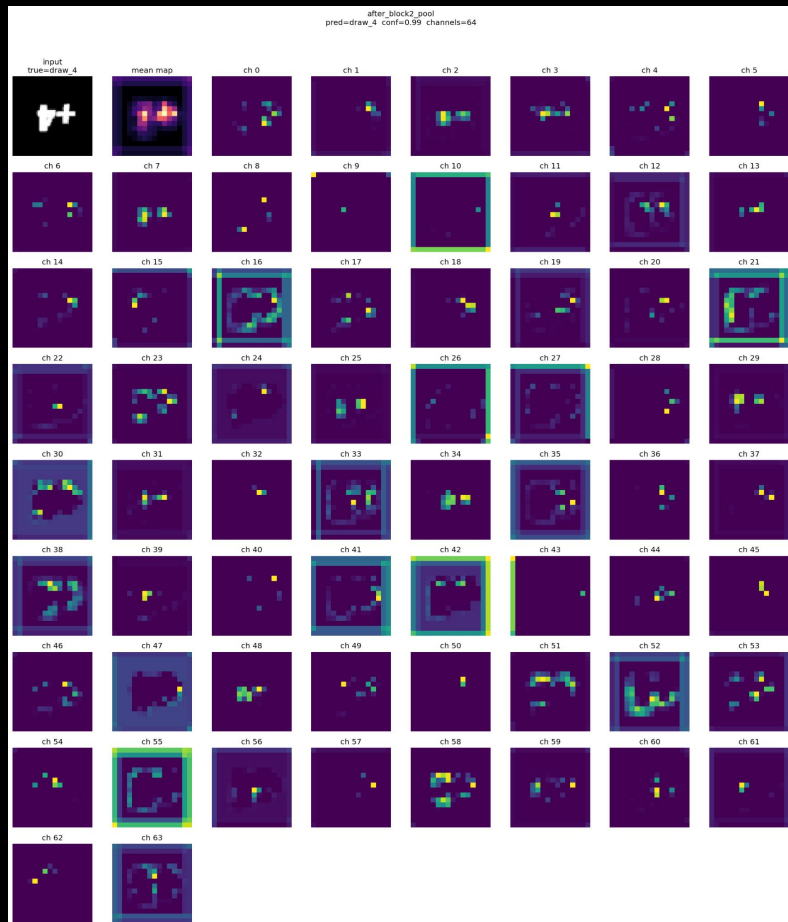
Classification: CNN Architecture



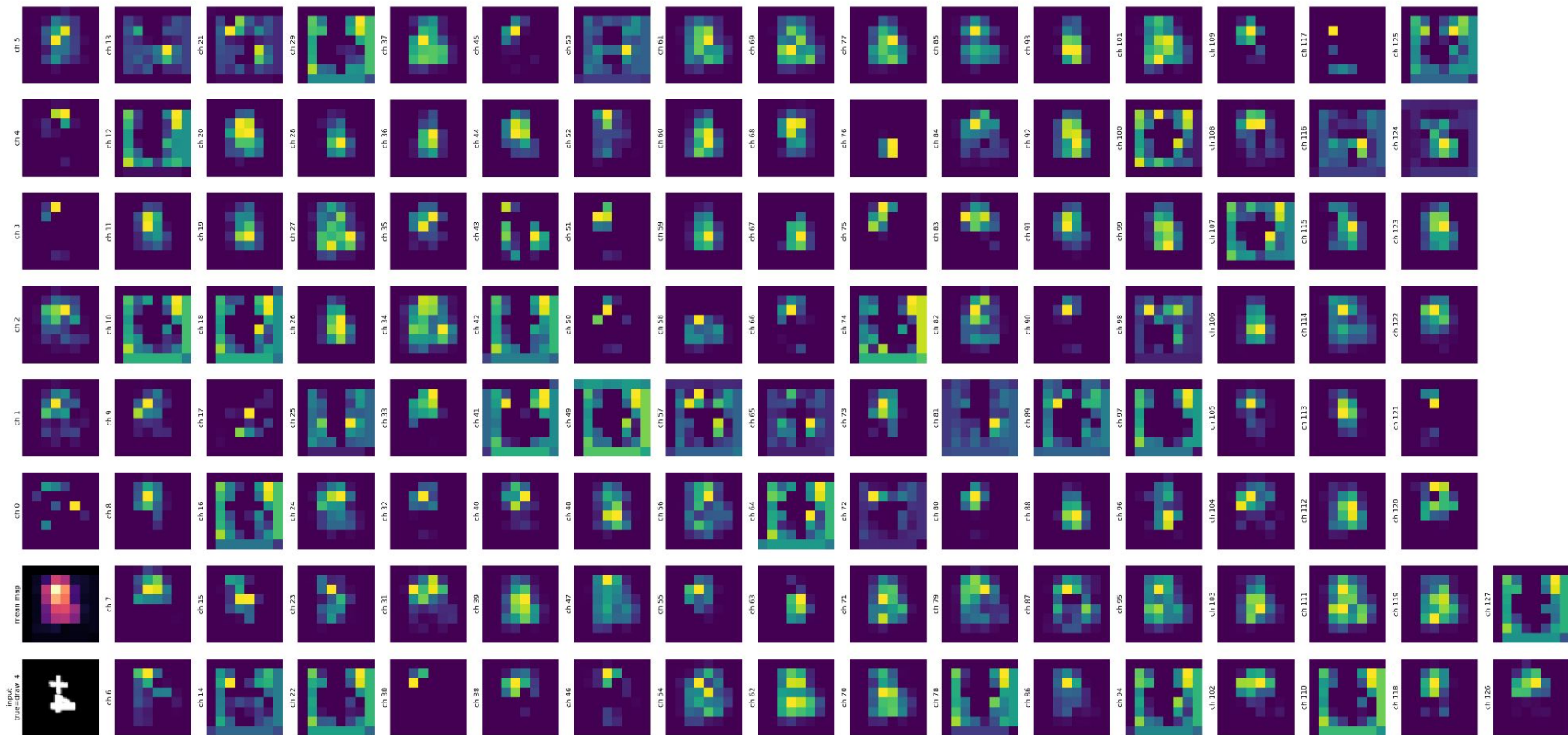
Classification: CNN Architecture



Classification: CNN Architecture



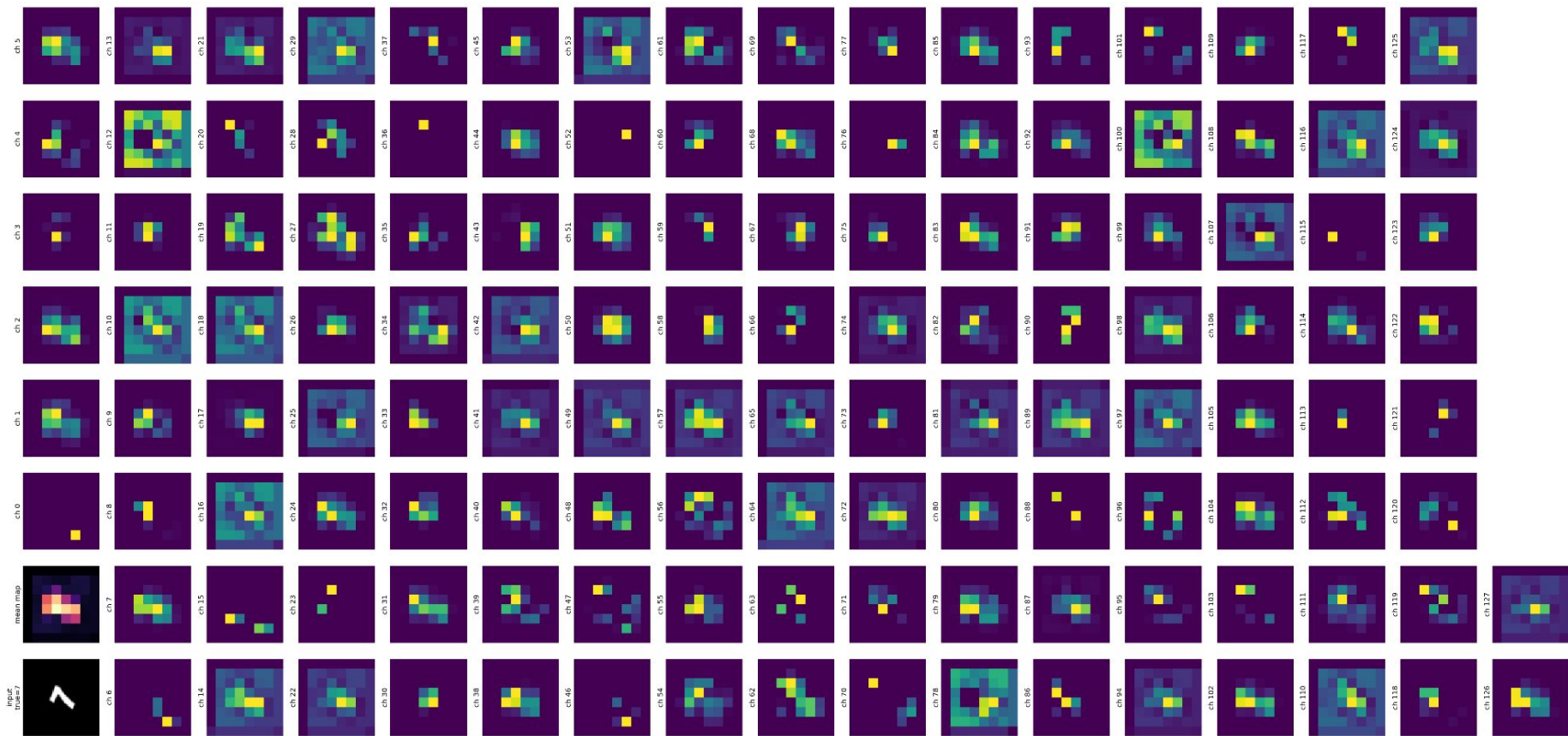
Classification: CNN Architecture



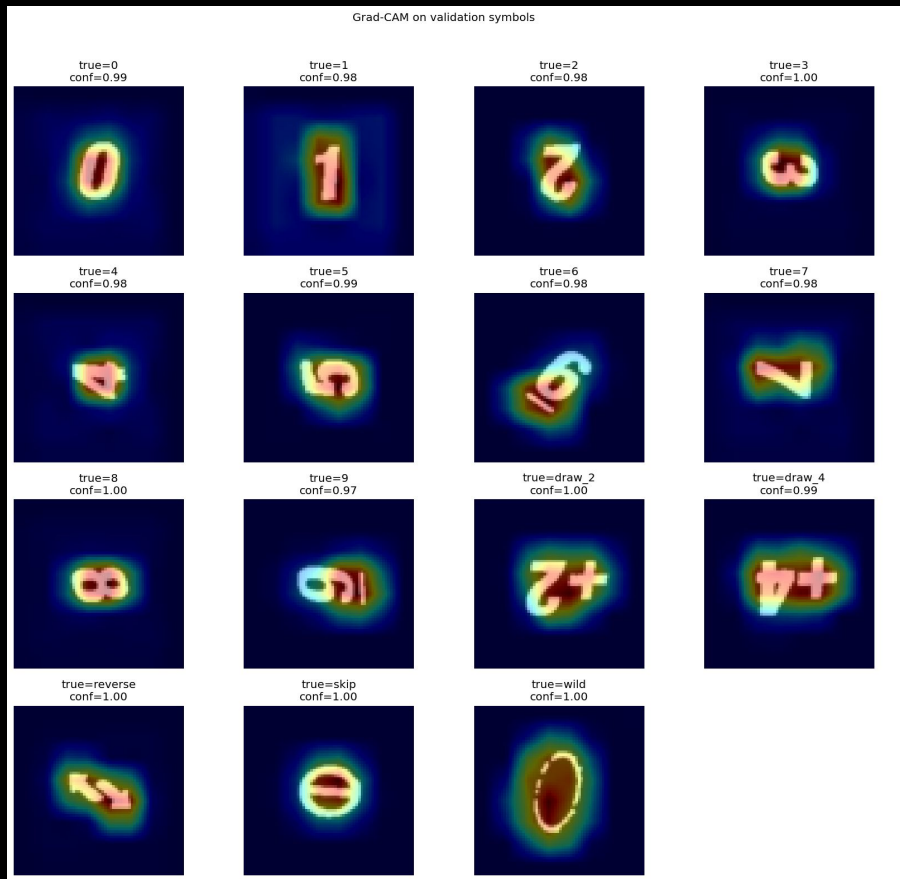
after_block3_pool
pred=draw_4 conf=0.99 channels=128

Classification: CNN Architecture

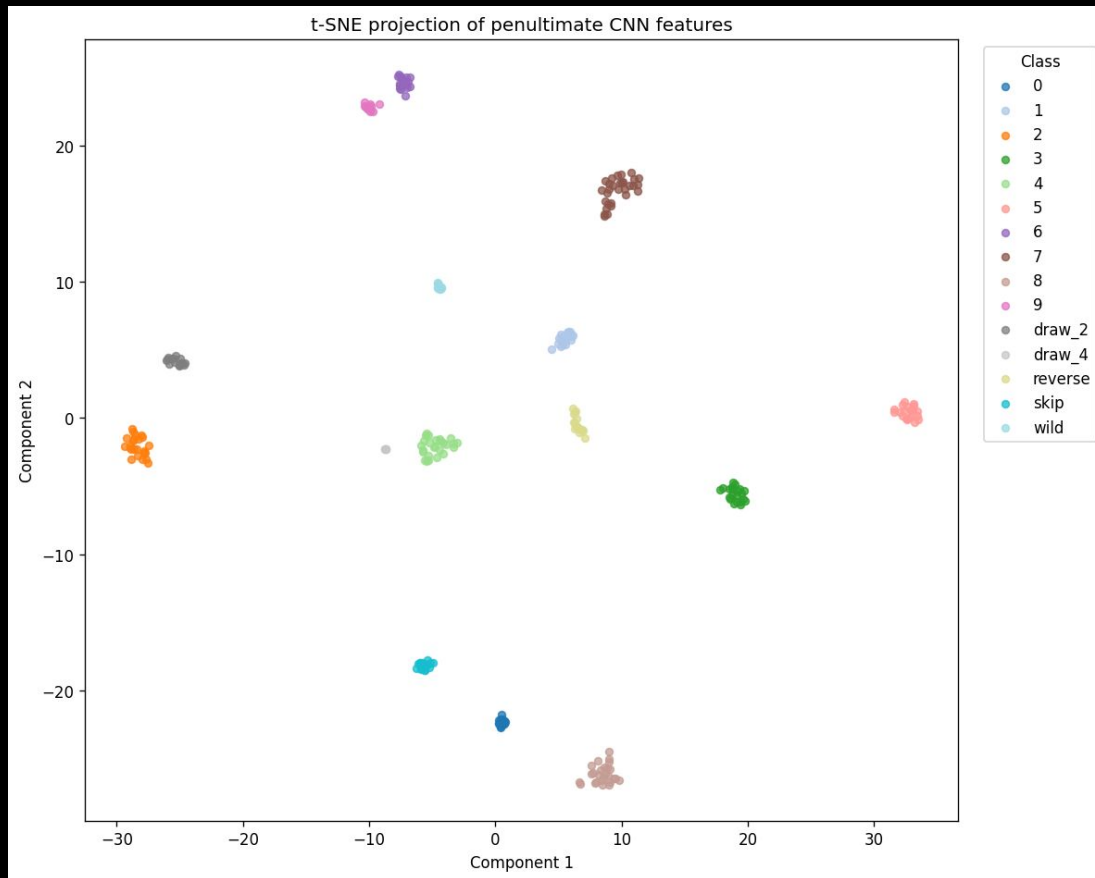
alex_kluc3_5_caf
pred=7 cor=0.91 channels=128



Classification: Gradient-weighted Class Activation Mapping

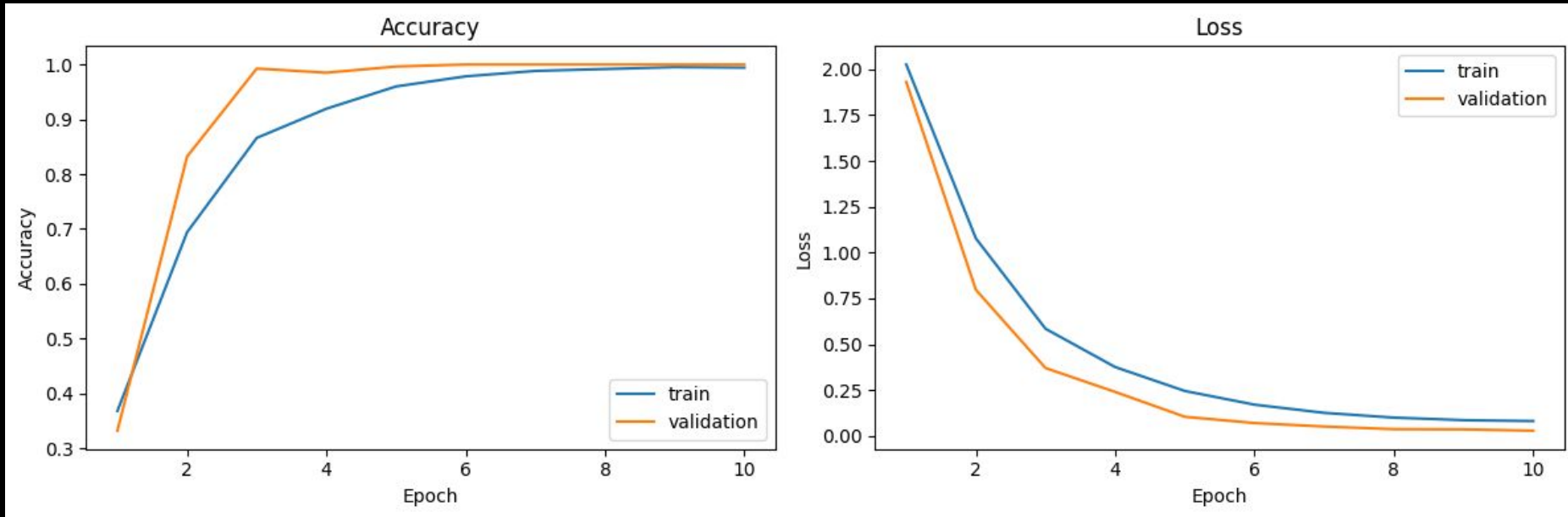


Classification: Feature vector projection



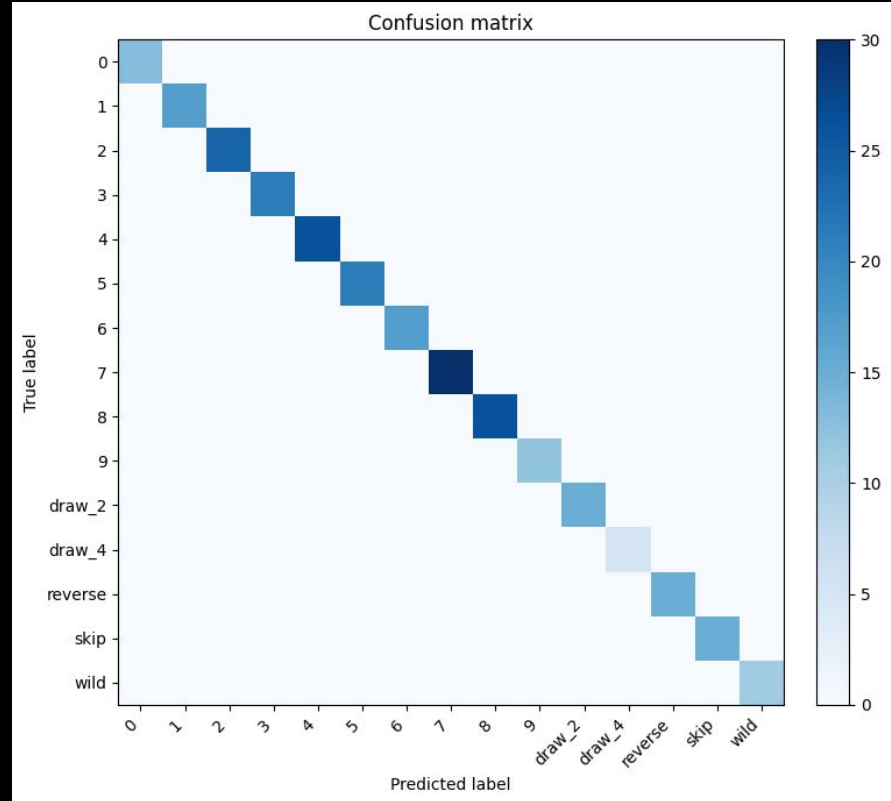
Classification: CNN Training & Validation

Cross-Entropy Loss

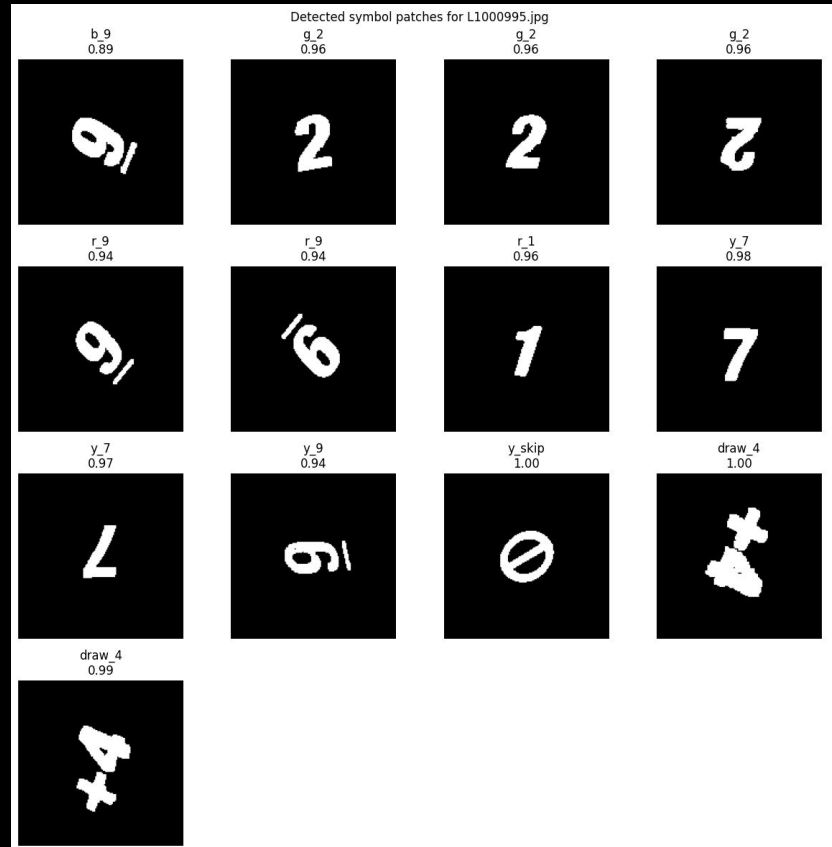


Classification: CNN Training & Validation

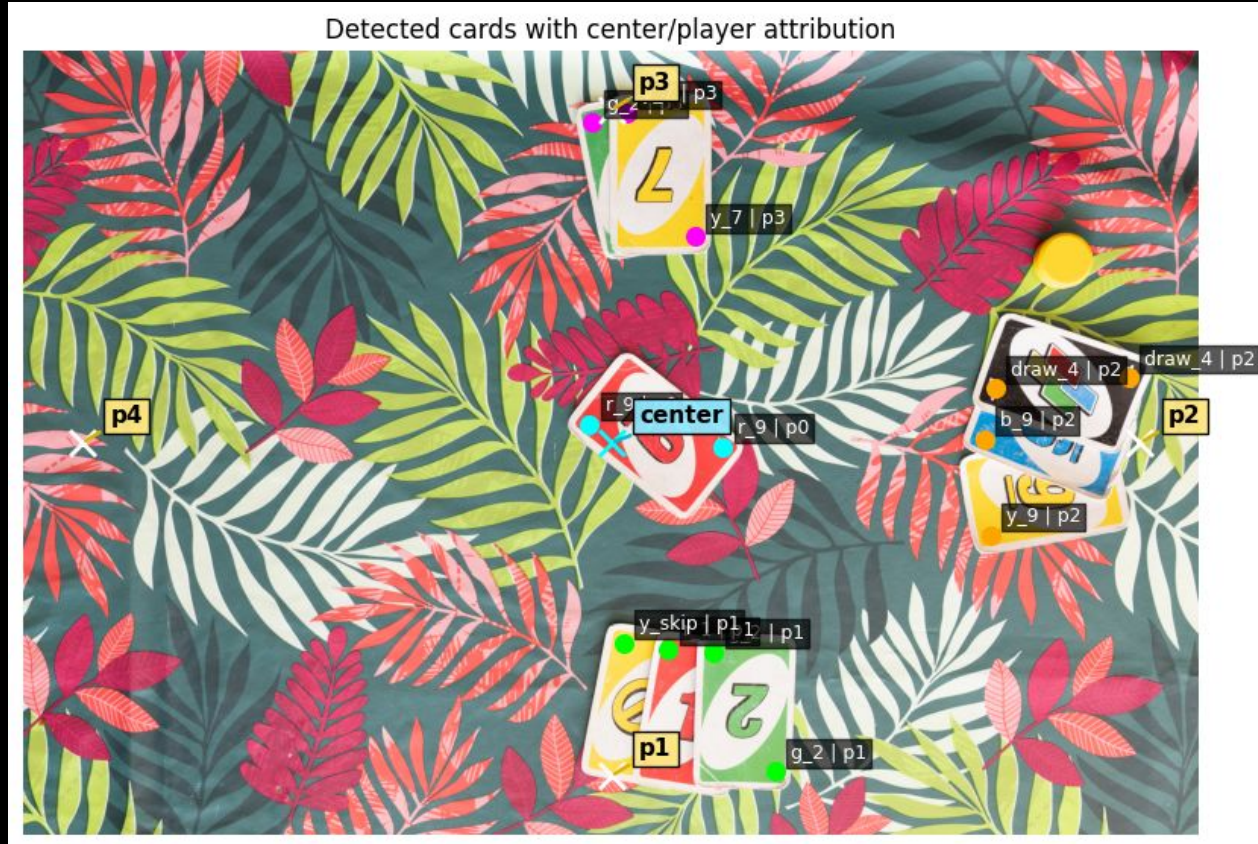
- F1 score : 1.0
- Accuracy : 1.0



Classification: CNN Training & Validation

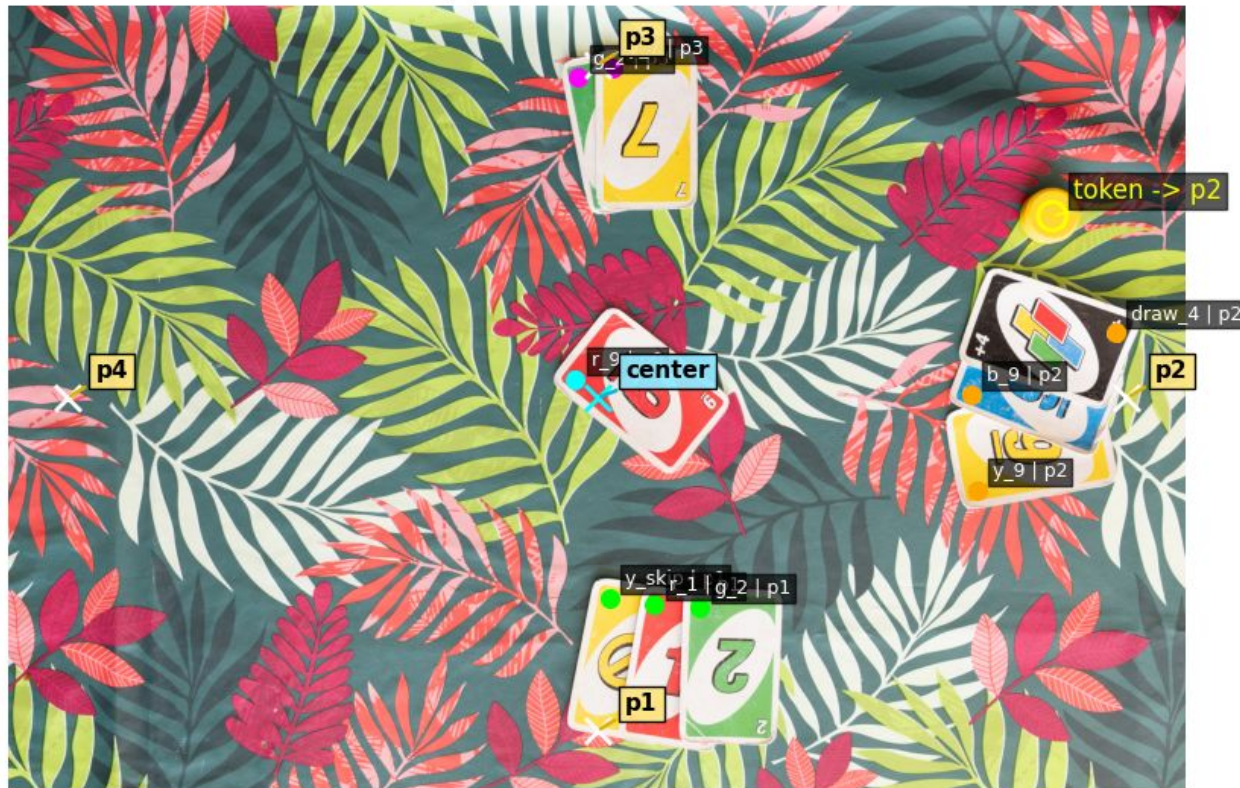


Post-processing: Symbol to player assignment



Post-processing: Final result

Full pipeline result on one frame





**THANK YOU FOR YOUR
ATTENTION**



Cross-Entropy Loss

For one training example with true class y

$$\mathcal{L} = -\log \left(\frac{e^{z_y}}{\sum_{j=1}^C e^{z_j}} \right)$$

Batch version

$$\mathcal{L}_{batch} = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{e^{z_{i,y_i}}}{\sum_{j=1}^C e^{z_{i,j}}} \right)$$

Symbols

C	number of classes = 15
z_j	raw model score for class j
z_y	score of the correct class

Training objective

Increase the correct class score and reduce competing class scores.